

Harsh Modi

Software Engineer | Full-Stack & AI Systems

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Summary

Full-stack Software Engineer with **3+ years of experience** shipping production software end-to-end – **React/TypeScript** interfaces, **Node.js** and **Python (FastAPI)** backends, and the infrastructure in between. Comfortable owning a feature from architecture through delivery, with hands-on experience building AI/LLM-powered systems: multi-provider model integration, retrieval pipelines, and interfaces for self-hosted open models. Pragmatic about trade-offs and focused on outcomes over any single tool or stack.

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL, HTML, CSS

Frontend: React.js, Next.js, React Hooks, responsive UI, accessibility

Backend: Node.js, Express.js, FastAPI, REST APIs, Socket.IO, PostgreSQL, Redis, Elasticsearch

AI/LLM: OpenAI API, Anthropic API, Gemini API, Groq, OpenRouter, Ollama, LangGraph, prompt evaluation, model comparison, token/cost tracking

Security/DevOps/Testing: AES-GCM, HKDF, SHA-256, Zod, Helmet, Docker, Docker Compose, Jenkins, Jest, Supertest, Postman, Git, Agile/Scrum

Professional Experience

Shoptaki | Software Engineer

USA | June 2025 – Present

- Built a secure, local-first messaging service using **Node.js**, **Express**, and **Socket.IO**, covering **20+ REST endpoints** for contacts, conversations, delivery state, and admin workflows.
- Implemented security controls across onboarding and message lifecycles with **Zod validation**, **Helmet**, rate limiting, **AES-GCM encryption**, per-message key derivation using **HKDF**, and salted **SHA-256** contact discovery.
- Designed an **in-memory TTL index** for contact hash matching, removing repeated storage scans and keeping lookup responses around **5–10 ms** in local load testing.
- Owned the architecture for **5 backend services** (Gateway, Auth, User, Master, Messaging), standardizing local development with **Docker Compose**, **Postman** collections, and runbooks so other engineers could run the stack consistently.

IBM | Software Developer

India | October 2021 – August 2023

- Developed enterprise **React** and **TypeScript** dashboards with reusable, component-driven tables, filters, forms, and detail screens across internal business workflows.
- Built and maintained **Node.js/Express** REST endpoints and **SQL/NoSQL** model updates, including request validation, predictable error handling, and integration test support.
- Improved dashboard performance by lazy-loading screens, trimming API payloads, and reducing page-load time by about **35%** across targeted dashboard views.
- Raised frontend quality through reusable components and accessibility fixes, helping dashboard pages reach **95+ Lighthouse accessibility** scores during QA review.

Projects

AI Incident Command Center | React | FastAPI (Python) | LangGraph | PostgreSQL | Elasticsearch | Redis | Docker

- Built an AI incident analysis dashboard (**React** frontend, **FastAPI** backend) that ingests logs, alerts, metrics, and synthetic incident chat transcripts to identify likely root cause, affected services, timeline of events, and recommended next actions.
- Designed a retrieval and correlation pipeline using PostgreSQL, Elasticsearch, and Redis to group related alerts, surface repeated failure patterns, and generate engineer-ready incident summaries.
- Implemented postmortem generation, Jira-style ticket drafts, and follow-up question prompts so teams can move from alert triage to resolution planning faster.

OpenPilot AI: Open-Source AI Developer Toolkit | Next.js | TypeScript | Node.js | PostgreSQL | Redis | Docker | OpenAI/Anthropic/Gemini/Ollama

- Built a full-stack AI developer platform (**Next.js/TypeScript** frontend, **Node.js** backend) for comparing LLM responses across OpenAI, Anthropic, Gemini, Groq, OpenRouter, and local Ollama models, with streaming chat, prompt history, and side-by-side evaluation.
- Implemented API key management, token usage tracking, model-level cost estimates, and saved prompt templates to help developers test prompts and monitor AI spend.
- Added a GitHub PR review module that analyzes pull-request diffs for security, performance, complexity, duplicate logic, and missing tests using configurable LLM providers.

Education

Master of Science in Computer Science, Pace University, USA

May 2025

Bachelor of Technology in Computer Science, Ahmedabad University, India

June 2023

Certifications

Front-End Apps with React – IBM

May 2023

Python Data Structures – University of Michigan

December 2021